Interpret line graphs

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: _________________

1. Read the model: A line graph rises from 18 to 46. What is the increase? Copy or complete the first step.

2. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

3. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

4. Miro says: Miro divides by the largest value instead of the number of values when finding the mean. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A line graph rises from 18 to 46. What is the increase? Copy or complete the first step.
   Answer 28.

2. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
   Answer 60 pupils.

3. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
   Answer 90 degrees.

4. Miro says: Miro divides by the largest value instead of the number of values when finding the mean. Is this correct? Explain.
   Answer Add every value, count how many values there are, then divide the total by that count.

Interpret line graphs

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A line graph rises from 18 to 46. What is the increase?

2. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

3. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

4. Solve this independently and show every stage: A line graph rises from 18 to 46. What is the increase?

5. Use a second method or representation for: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

6. Check this result using an inverse, estimate or known fact: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

7. Prove or explain your reasoning: Find the mean of 12, 15, 18, 11 and 19.

8. Identify and correct this misconception: Miro divides by the largest value instead of the number of values when finding the mean.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

Teacher answers and checking prompts

1. A line graph rises from 18 to 46. What is the increase?
   Answer 28.

2. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
   Answer 60 pupils.

3. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
   Answer 90 degrees.

4. Solve this independently and show every stage: A line graph rises from 18 to 46. What is the increase?
   Answer 28.

5. Use a second method or representation for: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
   Answer Expected result: 60 pupils. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
   Answer Expected result: 90 degrees. A valid check must be shown.

7. Prove or explain your reasoning: Find the mean of 12, 15, 18, 11 and 19.
   Answer 15. The explanation must show why the method works.

8. Identify and correct this misconception: Miro divides by the largest value instead of the number of values when finding the mean.
   Answer Add every value, count how many values there are, then divide the total by that count.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Interpret line graphs

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Find the mean of 12, 15, 18, 11 and 19.

2. Prove the answer to this question, rather than only calculating it: A line graph rises from 18 to 46. What is the increase?

3. Solve this in two different ways and compare them: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

4. Work backwards from the answer to reconstruct the question: 90 degrees.

5. Explain why this reasoning fails: Miro divides by the largest value instead of the number of values when finding the mean.

6. Create a new example that tests the same idea as: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the mean of 12, 15, 18, 11 and 19.
   Answer 15. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A line graph rises from 18 to 46. What is the increase?
   Answer Expected result: 28. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
   Answer Expected result: 60 pupils. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 90 degrees.
   Answer One valid reconstruction is: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro divides by the largest value instead of the number of values when finding the mean.
   Answer Add every value, count how many values there are, then divide the total by that count.

6. Create a new example that tests the same idea as: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.